# University Clustering and Imputation App

K-means segments of 1,302 U.S. universities, served through a Flask web app

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| 4 | 93% | 34% |
|---|---|---|
| university segments | cluster-sector agreement | lower tuition-imputation error vs. mean |

## 1. Background and question

**What natural groups of U.S. universities exist, and can those groups fill in a university's missing data?**

Universities differ on size, cost, selectivity and outcomes, and many records are incomplete. Grouping similar universities makes the landscape easier to understand and gives a principled way to estimate missing values from a school's peers.

## 2. Data

- 1,302 U.S. colleges and universities, 17 numeric measures: applications, acceptances, enrolment, student quality, tuition and costs, faculty, graduation rate
- Name, state and public/private kept aside to check the clusters
- 831 universities (64%) have at least one missing value; board, room and fees are missing most often

## 3. Method

- Mean-impute for fitting, standardise all 17 measures, K-means with k from 2 to 10 compared by elbow and silhouette; k = 4 chosen for interpretability
- Cluster profiles in original units; clusters named from their profiles (sector, size, selectivity)
- Validation against sector and state, which were not used in clustering
- Imputation: distance to each cluster centre using only the values a university has, in standardised units; gaps filled with the nearest centre's averages; accuracy tested by hiding 150 known values per field
- Flask web app with sign-up, hashed-password login and a JSON imputation endpoint; pytest tests for login and imputation

## 4. Project walkthrough

**The problem.** Find natural groups among 1,302 U.S. universities, describe them, and use them to fill in missing data through a small web app.

**Preparing the data.** Seventeen numeric measures were mean-imputed for fitting and standardised, so that columns measured in thousands (applications, tuition) do not dominate columns measured in percent.

**Choosing k.** The silhouette score peaks at 2-3 clusters and the elbow has no sharp bend, so I chose k = 4 for interpretability: it separates two public and two private groups, which k = 3 merges.

**Checking the clusters.** Public/private status was never given to the model, yet 93% of universities land in a cluster matching their sector, a strong sign the clusters are real.

**Imputation.** For a university with gaps, I measure its distance to each cluster centre using only the values it has, then fill each gap with the nearest centre's average. Hiding 150 known values per field showed 13-34% lower error than filling with the overall mean.

**The web app.** A Flask app with sign-up, hashed-password login and an imputation endpoint. While rebuilding it I found and fixed a serious bug: the original login accepted any password for an existing user. Tests now check this.

## 5. Results

| Field (150 hidden values) | Cluster MAE | Mean-fill MAE | Improvement |
|---|---|---|---|
| Graduation rate (%) | 12.5 | 16.5 | 24% |
| % new students from top 10% | 12.7 | 14.5 | 13% |
| Out-of-state tuition ($) | 2,308 | 3,485 | 34% |

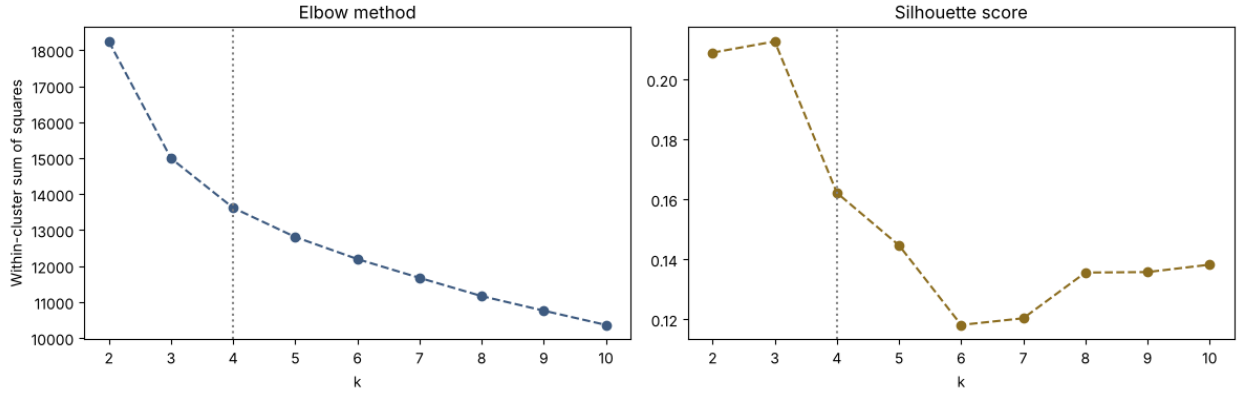



*Figure: Choosing k: elbow (left) and silhouette score (right)*

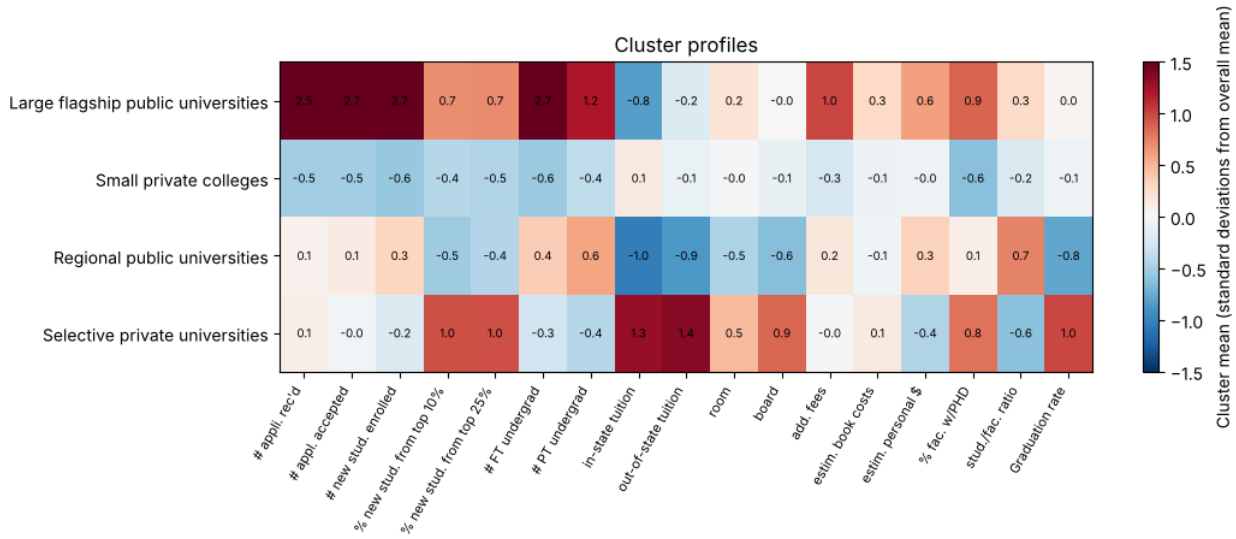



*Figure: Cluster profiles: how far each segment sits from the overall mean on every measure*

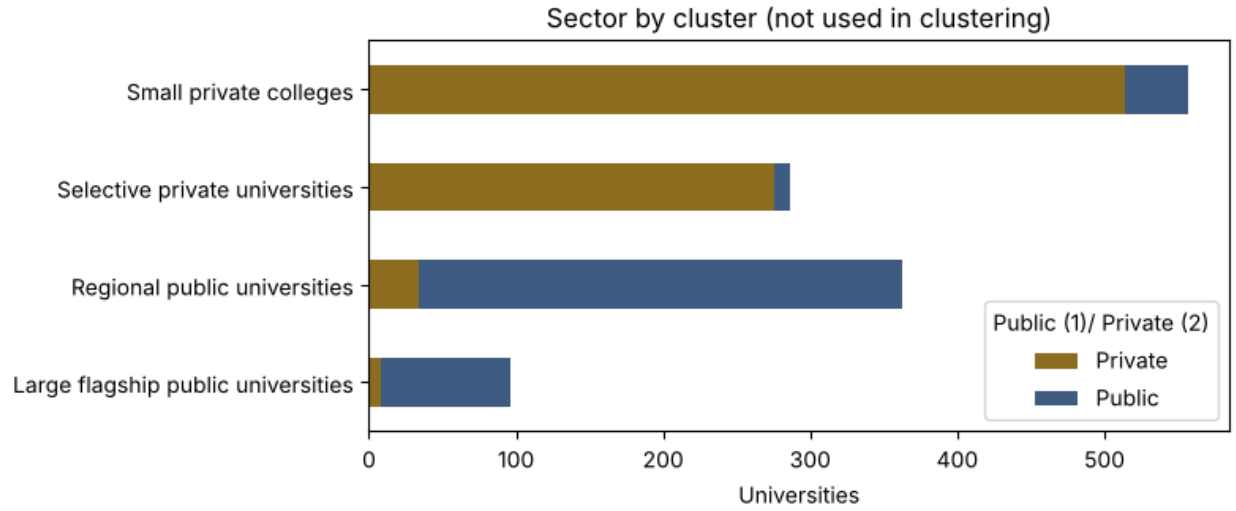



*Figure: Public vs. private universities in each cluster (sector was not used in clustering)*

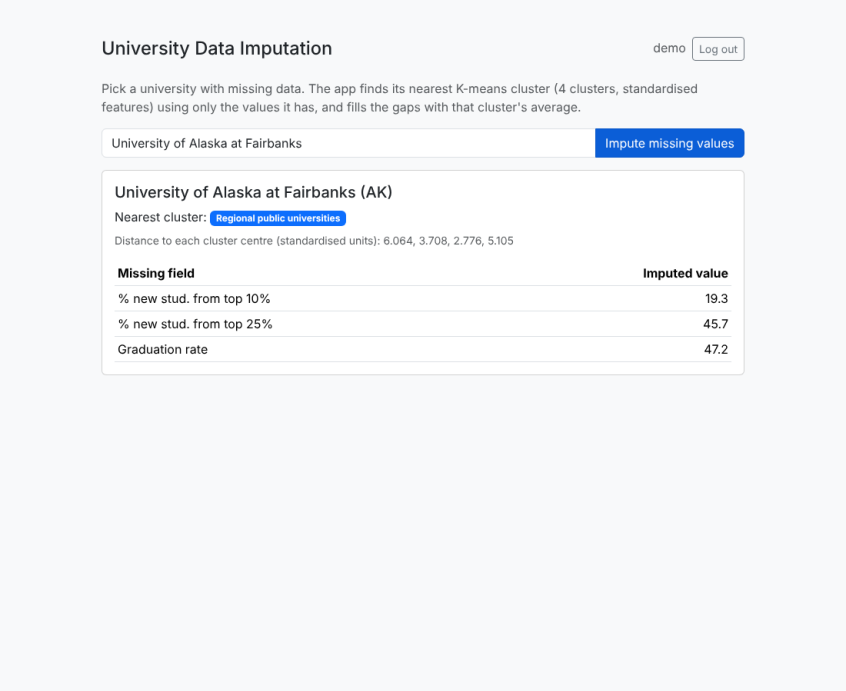



*Figure: The Flask app imputing missing values for the University of Alaska Fairbanks*

## 6. Key findings

- Four segments: large flagship public universities (96; ~16,000 undergraduates, selective), regional public universities (363; lowest tuition and 46% graduation), small private colleges (557; ~1,200 students) and selective private universities (286; 79% graduation, highest tuition).
- Sector was never used in clustering, yet 93% of universities fall in a cluster matching their sector; selective private universities concentrate in NY, PA, MA and CA.
- Cluster-based imputation beat mean-filling on every field tested: 24% lower error for graduation rate and 34% for out-of-state tuition.
- Fixed four bugs in the original app: login never checked passwords, the app clustered unstandardised data with 3 clusters while the analysis used 4 standardised ones, broken redirects, and a hard-coded secret key.

## 7. Why these choices

**Standardising before K-means.** K-means uses Euclidean distance, so unscaled tuition (thousands of dollars) would swamp percentages. The original app skipped this; the corrected version shares one standardised model with the notebook.

**Distance on available columns only.** A university missing graduation rate can still be placed using its other 16 measures, so imputation uses whatever is known.

**Testing the imputation.** Hiding known values and measuring the error turns 'it fills gaps' into a measurable claim.

## 8. Key terms explained

| | |
|---|---|
| **K-means** | Groups data into k clusters by repeatedly assigning points to the nearest centre and moving each centre to its points' average. |
| **Silhouette score** | How much closer points are to their own cluster than to the next one, from -1 to 1; higher means better-separated clusters. |
| **Standardisation** | Rescaling each measure to mean 0 and standard deviation 1 so all measures count equally in distances. |
| **Password hashing** | Storing a one-way scrambled version of a password so the real password is never saved. |

## Tools and skills

Python, pandas, scikit-learn (K-means, StandardScaler, silhouette), Flask, SQLite, Werkzeug password hashing, pytest, Bootstrap, matplotlib

Code: university_clustering.ipynb in this folder reproduces every number in this report.